



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester MCA (CBCS) Degree Examinations

January 2024

22MCA343 – BLOCK CHAIN TECHNOLOGY

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit

		Unit – I			
		Marks	BT*	CO*	PO*
1.	a) Define Blockchain. Discuss the working of Blockchain.	10	L*2	1	1
	b) List and explain Myths about Bitcoin.	10	L2	1	1
2.	a) Describe the usage of private and public keys in Blockchain.	10	L3	1	1
	b) List and explain the advantage and disadvantage of Blockchain.	10	L2	1	1
		Unit – II			
3.	a) With help of a neat diagram, explain Blockchain-architecture.	10	L2	2	1
	b) Discuss the uses of shared database. List the difference between Blockchain and Shared Database.	10	L2	2	1
4.	a) Illustrate the different types Cryptocurrencies and its applications.	10	L2	2	1
	b) Discuss different variants of Blockchain.	10	L2	2	1
		Unit – III			
5.	a) Explain the concept of Merkle Tree.	10	L2	3	1
	b) Define double Spending. List and explain the types of double spending attacks and solutions to prevent double spending.	10	L2	3	1
6.	a) With the help of an example, explain how Bitcoin handles double spending.	10	L2	3	1
	b) Explain the concept of Resolving conflicts in bitcoin network.	10	L2	3	1
		Unit – IV			
7.	a) Explain the concept of Converting Bitcoins to Fiat Currency.	10	L2	4	1
	b) With examples, explain the process of sending and receiving Bitcoin.	10	L2	4	1
8.	a) Describe Bitcoin Forks and SegWit in detail.	10	L2	4	1
	b) What is bitcoin? Explain the key concepts of bitcoin.	10	L2	4	1
		Unit – V			
9.	a) Discuss the key concepts of Ethereum. Compare Bitcoin and Ethereum.	10	L2	5	1
	b) Explain the working of Smart Contracts.	10	L2	5	1
10.	a) Summarize the features of Ethereum.	10	L2	5	1
	b) What are the Capabilities of Smart Contracts? Describe.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
